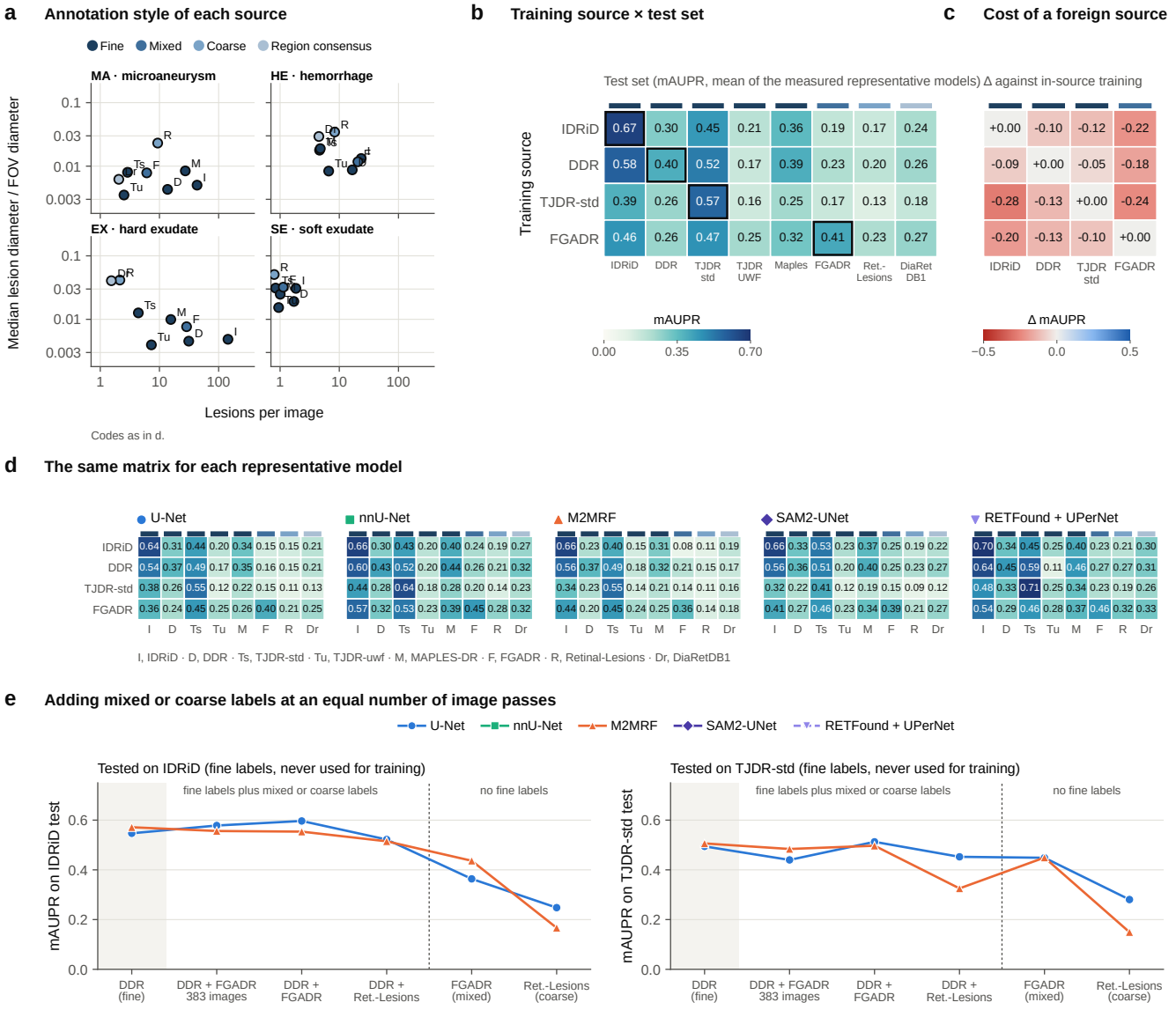
a
Annotation style of each source
Fine
Mixed
Coarse
Region consensus
MA · microaneurysm
HE · hemorrhage
EX · hard exudate
SE · soft exudate
Median lesion diameter / FOV diameter
Lesions per image
Codes as in d.
b
Training source × test set
Test set (mAUPR, mean of the measured representative models)
Training source
IDRiD
DDR
TJDR-std
FGADR
IDRiD
DDR
TJDR std
TJDR UWF
Maples
FGADR
Ret.-Lesions
DiaRet DB1
mAUPR
0.00
0.35
0.70
c
Cost of a foreign source
Δ against in-source training
Δ mAUPR
−0.5
0.0
0.5
d
The same matrix for each representative model
U-Net
nnU-Net
M2MRF
SAM2-UNet
RETFound + UPerNet
I, IDRiD · D, DDR · Ts, TJDR-std · Tu, TJDR-uwf · M, MAPLES-DR · F, FGADR · R, Retinal-Lesions · Dr, DiaRetDB1
e
Adding mixed or coarse labels at an equal number of image passes
U-Net
nnU-Net
M2MRF
SAM2-UNet
RETFound + UPerNet
Tested on IDRiD (fine labels, never used for training)
mAUPR on IDRiD test
fine labels plus mixed or coarse labels
no fine labels
DDR (fine)
DDR + FGADR 383 images
DDR + FGADR
DDR + Ret.-Lesions
FGADR (mixed)
Ret.-Lesions (coarse)
Tested on TJDR-std (fine labels, never used for training)
mAUPR on TJDR-std test